

J5 Vector problems

Name: _____

Class: _____

Date: _____

Time: **410 minutes**

Marks: **336 marks**

Comments:

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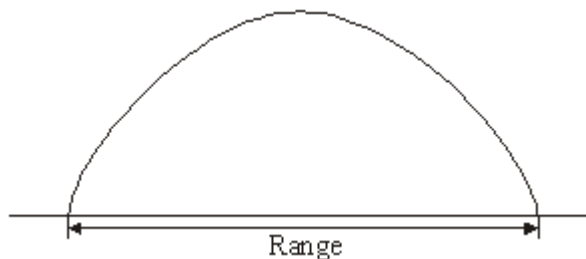
Q1.

A ball is hit from horizontal ground with velocity $(10\mathbf{i} + 24.5\mathbf{j}) \text{ m s}^{-1}$ where the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertically upwards respectively.

- (a) State two assumptions that you should make about the ball in order to make predictions about its motion.

(2)

- (b) The path of the ball is shown in the diagram.



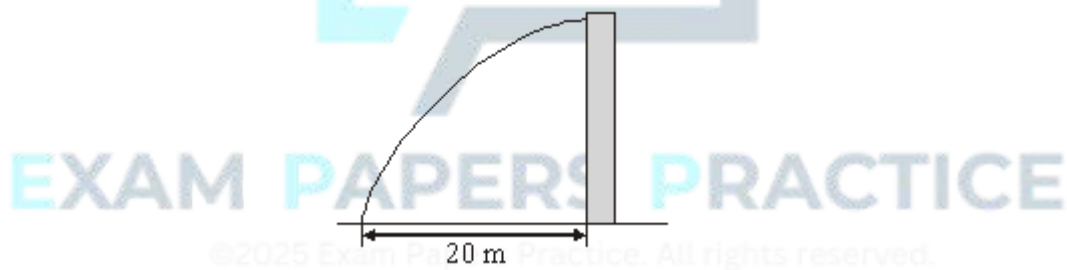
- (i) Show that the time of flight of the ball is 5 seconds.

(4)

- (ii) Find the range of the ball.

(2)

- (c) In fact the ball hits a vertical wall that is 20 metres from the initial position of the ball.



Find the height of the ball when it hits the wall.

(4)

- (d) If a heavier ball were projected in the same way, would your answers to part (b) of this question change? Explain why.

(2)

(Total 14 marks)

Q2.

A particle P , of mass 0.25 kg , moves in a horizontal plane so that at time t seconds its velocity $\mathbf{v} \text{ m s}^{-1}$ is given by

$$\mathbf{v} = 8 \sin t \mathbf{i} + 4 \cos t \mathbf{j}$$

- (a) Find the force, $\mathbf{F}$ newtons, acting on P at time t . (3)

- (b) (i) Show that the magnitude of $\mathbf{F}$ at time t is

$$\sqrt{(3 \cos^2 t + 1)}$$

(3)

- (ii) Hence state the range of values of the magnitude of $\mathbf{F}$.

(2)

(Total 8 marks)

Q3.

A particle P moves in a horizontal plane so that at time t seconds after the motion begins its position vector $\mathbf{r}$ metres is given by

$$\mathbf{r} = \sin 2t \mathbf{i} + 6t \mathbf{j}$$

- (a) (i) Find the velocity of P at time t . (2)

- (ii) Show that the speed of P at time t is

$$\sqrt{(4 \cos^2 2t + 36)}$$

(3)

- (iii) Find the first value of t for which the speed of P has its minimum value.

(2)

- (b) (i) The mass of P is 0.25 kg. Find the force, $\mathbf{F}$ newtons, acting on P at time t . (3)

- (ii) State the direction of $\mathbf{F}$ and explain why its magnitude never exceeds one newton.

(2)

(Total 12 marks)

Q4.

A particle P , of mass 0.5 kg, moves under the action of a force $\mathbf{F}$ newtons, where $\mathbf{F} = 2t\mathbf{i} + 1.5\mathbf{j}$ at time t seconds.

- (a) Find the acceleration of P at time t . (1)

- (b) The particle P starts its motion from rest.

Find the velocity of P at time t .

(2)

(c) The particle P starts its motion at the point O .

(i) Find the position vector of P relative to O at time t .

(3)

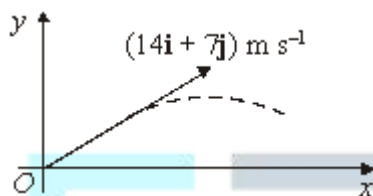
(ii) Find the distance of P from O when it is moving parallel to the vector $\mathbf{i} + \mathbf{j}$.

(5)

(Total 11 marks)

Q5.

Jonny is practising kicking a rugby ball. He kicks the ball from a point O on horizontal ground with velocity $14\mathbf{i} + 7\mathbf{j} \text{ m s}^{-1}$. The vectors $\mathbf{i}$ and $\mathbf{j}$ are in the directions of axes Ox and Oy , and Oy is vertically upwards.



(a) (i) Find the coordinates of the ball t seconds after it has been kicked.

(3)

(ii) Show that the equation of the path of the ball is

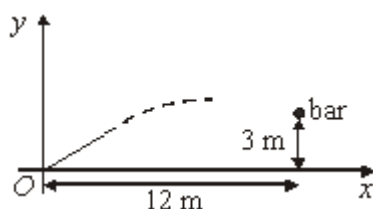
$$y = \frac{20x - x^2}{40}$$

(3)

(iii) Find the horizontal range of the ball.

(2)

(b) The ball moves towards a horizontal bar which is 3 metres above the ground and at a horizontal distance of 12 metres from O , as shown in the diagram.



Determine whether the ball passes above or below the bar.

(3)

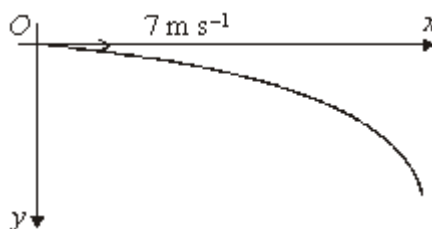
(c) State one modelling assumption that you have used in answering this question.

(1)

(Total 12 marks)

Q6.

Elaine kicks a ball off a cliff top. She kicks the ball from a point O and it subsequently moves in a vertical plane with respect to axes which are horizontal and vertically downwards, as shown in the diagram.



The initial velocity of the ball is horizontal and of magnitude 7 m s^{-1} .

- (a) Find the coordinates of the ball t seconds after it has been kicked.

(3)

- (b) Show that the equation of the path of the ball is

$$y = \frac{x^2}{10}.$$

(2)

The ball subsequently lands on a horizontal beach. The point A on the beach is vertically below O and the ball lands at the point B on the beach, as shown in the diagram.



- (c) The distance OA is 8.1 metres.

Find the distance AB .

(2)

- (d) Find the speed of the ball as it reaches B .

(5)

(Total 12 marks)

Q7.

A particle P moves so that at time t seconds its position vector, $\mathbf{r}$ metres, is given by

$$\mathbf{r} = 6t\mathbf{i} + t^2\mathbf{j}.$$

- (a) Find the velocity of P at time t . (2)
- (b) Find an expression for the speed of P at time t . (2)
- (c) Find the value of t when P moves with speed $6\sqrt{2} \text{ m s}^{-1}$. (2)
- (Total 6 marks)

Q8.

A particle P moves so that at time t it has velocity

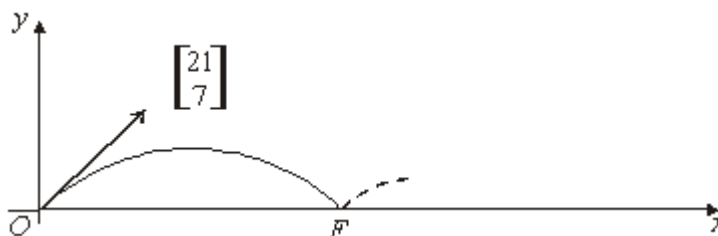
$$\mathbf{v} = 4\mathbf{i} - 2t\mathbf{j}, \quad t \geq 0.$$

When $t = 0$, the particle P has position vector $8\mathbf{j}$.

- (a) Find the position vector of P at time t . (4)
- (b) Find the position vector of P when $t = 2$. (1)
- (c) Find the time when the position vector of P is in the direction of the vector $\mathbf{i}$. (2)
- (Total 7 marks)

Q9.

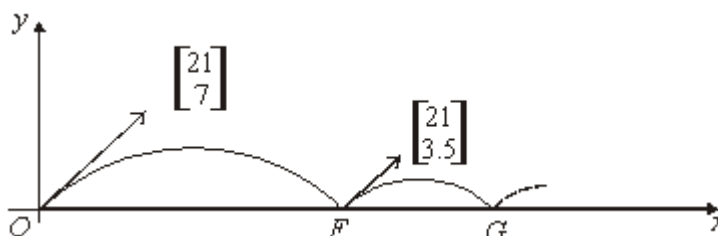
A ball is kicked from a point O on horizontal ground with velocity $\begin{bmatrix} 21 \\ 7 \end{bmatrix} \text{ m s}^{-1}$.
 The ball subsequently hits the ground at a point F , as shown in the diagram.



- (a) (i) Find the time between the ball being kicked and reaching F . (3)
- (ii) Find the distance OF . (2)

- (b) The ball rebounds from F with velocity $\begin{bmatrix} 21 \\ 3.5 \end{bmatrix}$ m s⁻¹. It subsequently hits the ground again at

G and then rebounds, as shown in the diagram below.



Find the distance FG .

(4)

- (c) After bouncing at G , the ball rebounds with velocity $\begin{bmatrix} 21 \\ 1.75 \end{bmatrix}$ m s⁻¹, so that the vertical component of the velocity is halved again, and then subsequently bounces next at H .

Find the total distance OH .

(3)

(Total 12 marks)

Q10.

A particle P moves so that at time t seconds its velocity $\mathbf{v}$ m s⁻¹ is

$$\mathbf{v} = 2t\mathbf{i} + 4\mathbf{j}, \quad t \geq 0.$$

- (a) At time $t = 0$, the particle is at the point with position vector $2\mathbf{j}$ metres.

Find the position vector of P at time t .

(4)

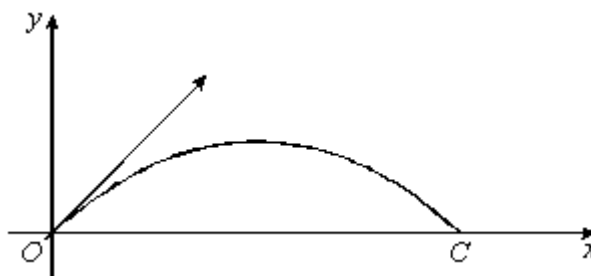
- (b) At times $t = 2$ and $t = 4$, the particle passes through the points C and D respectively.

Find the vector $\overrightarrow{CD}$.

(4)

(Total 8 marks)

Q11.



Oliver throws a ball from point O , with velocity $\begin{bmatrix} 7 \\ 7 \end{bmatrix}$ m s⁻¹. The ball is subsequently caught by

Chris at a point C which is on the same horizontal level as O , as shown in the diagram. During the motion from O to C the ball moves in a vertical plane.

- (a) Taking O as the origin, find the position vector of the ball t seconds after being thrown.

(3)

- (b) Show that the equation of the path of the ball is

$$y = x - 0.1x^2,$$

where (x, y) are the coordinates at time t of the ball relative to the origin O .

(4)

- (c) A horizontal wire, which is part of a fence, cuts the plane of motion of the ball at the point $(4, 2)$.

Determine whether the ball passes above or below the wire.

(3)

- (d) Find the distance OC .

(4)

- (e) Find the greatest speed of the ball during its motion from O to C .

(2)

(Total 16 marks)

Q12.

At time t , the position vector of a particle Q is

$$\mathbf{r} = (t^2 - 6t + 4)\mathbf{i} + \frac{1}{3}t^3\mathbf{j}.$$

- (a) Find the velocity of Q at time t .

(3)

- (b) Find the value of t when Q is moving parallel to the vector $\mathbf{j}$.

(2)

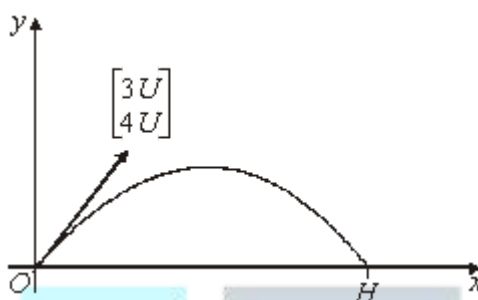
- (c) Find the acceleration of Q and state, with a reason, whether or not this acceleration is constant.

(2)

(Total 7 marks)

Q13.

George throws a ball from a point O , with velocity $\begin{bmatrix} 3U \\ 4U \end{bmatrix}$. The ball subsequently lands at a point H , which is at the same horizontal level as O , as shown in the diagram.



- (a) Show that the time taken by the ball to travel from O to H is $\frac{8U}{g}$. (3)
- (b) Find, in terms of g and U , the distance OH . (2)
- (c) Find, in terms of U , the initial speed of the ball. (2)
- (d) Find, in terms of g and U , the two times during the flight from O to H when the ball is moving with speed $\sqrt{18U}$. (6)

(Total 13 marks)

Q14.

A particle moves with constant acceleration in a horizontal plane. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the plane. At time $t = 0$, the particle is at rest at the origin. At time $t = 4$ seconds, the position vector of the particle is $(-16\mathbf{i} + 16\mathbf{j})$ metres.

- (a) Show that the acceleration of the particle is $(-2\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-2}$. (4)
- (b) Find the speed of the particle when $t = 5$.

(4)

- (c) The mass of the particle is 3 kg. Two horizontal forces **P** and **Q** act on the particle. If $\mathbf{P} = 20\mathbf{i} - 10\mathbf{j}$, find **Q**.

(4)

(Total 12 marks)

Q15.

A particle moves so that, at time t seconds, its position vector, $\mathbf{r}$ metres, is given by

$$\mathbf{r} = 4e^{-t}\mathbf{i} + (6t + 3e^{-t})\mathbf{j}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

- (a) Show that the velocity of the particle is $(-4\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$ when $t = 0$. (3)
- (b) Find an expression for the acceleration of the particle at time t . (2)
- (c) Find the magnitude of the acceleration of the particle when $t = 0$. (2)
- (d) Describe what happens to the velocity of the particle as t becomes large. (2)

(Total 9 marks)

Q16.

A particle moves with constant acceleration $(4\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-2}$. At time $t = 0$, the particle is at the origin and has velocity $(3\mathbf{i} - 10\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Show that the velocity of the particle at time t seconds is $(3 + 4t)\mathbf{i} + (2t - 10)\mathbf{j}$. (2)
- (b) Find the time when the particle is travelling due east. (2)
- (c) (i) Find the position vector of the particle when $t = 10$. (3)
- (ii) Find the velocity of the particle when $t = 10$. (1)
- (d) The particle stops accelerating when $t = 10$ and then moves with constant velocity. Find the distance of the particle from the origin after it has been travelling with this

constant velocity for a further 10 seconds.

(6)

(Total 14 marks)

Q17.

A particle is initially at the origin and has velocity $(4\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$. It moves with constant acceleration, and 10 seconds later has velocity $(5\mathbf{i} - 2\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Show that the acceleration of the particle is $(0.1\mathbf{i} - 0.5\mathbf{j}) \text{ m s}^{-2}$.

(3)

- (b) Find an expression for the position vector of the particle at time t seconds.

(2)

- (c) Find the time when the particle is due east of the origin.

(4)

- (d) Find the time when the particle is moving due east.

(4)

(Total 13 marks)

Q18.

At time $t = 0$, a boat has velocity $(2\mathbf{i} - 3\mathbf{j}) \text{ m s}^{-1}$ and is at the origin. After moving with a constant acceleration of $\mathbf{a} \text{ m s}^{-2}$ for 20 seconds, its position vector is $(60\mathbf{i} + 20\mathbf{j})$ metres. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find $\mathbf{a}$.

(3)

- (b) Find the velocity of the boat at time t seconds.

(2)

- (c) Find the speed of the boat when it is travelling south east.

(5)

(Total 10 marks)

Q19.

A particle is initially at rest at the point A with position vector $(9\mathbf{i} + 10\mathbf{j})$ metres, with respect to an origin O . It moves with constant acceleration, and 10 seconds later is at the point B with position vector $(19\mathbf{i} - 25\mathbf{j})$ metres. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

- (a) Show that the acceleration of the particle is $(0.2\mathbf{i} - 0.7\mathbf{j}) \text{ m s}^{-2}$.

(4)

- (b) Find the speed of the particle at B .

(4)

- (c) Verify that the particle passes through the point with position vector $(15.4\mathbf{i} - 12.4\mathbf{j})$ m. (4)
- (d) Draw a diagram to show the path of the particle as it moves from A to B . You should include the unit vectors $\mathbf{i}$ and $\mathbf{j}$ on your diagram. (2)

(Total 14 marks)

Q20.

A particle moves in a horizontal plane. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

Initially the particle is at the origin and has velocity $(5\mathbf{i} - 2\mathbf{j})$ m s⁻¹. It moves with constant acceleration. At time $t = 10$ it has position vector $(40\mathbf{i} - 15\mathbf{j})$ metres.

- (a) Show that the velocity of the particle is $(3\mathbf{i} - \mathbf{j})$ m s⁻¹ when $t = 10$. (4)
- (b) Find the acceleration of the particle. (3)
- (c) The mass of the particle is 15 kg. Find the magnitude of the resultant force acting on the particle. (4)

(Total 11 marks)

Q21.

A particle of mass 4 kg moves on a smooth horizontal plane. It is initially at rest at the origin. A force $\mathbf{F} = (8\mathbf{i} - 12\mathbf{j})$ N acts on the particle for 20 seconds. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

- (a) (i) Find the acceleration of the particle. (2)
- (ii) Find the velocity of the particle at the end of the 20 second period. (2)
- (iii) Find the position of the particle at the end of the 20 second period. (3)
- (b) At the end of the 20 second period, the force $\mathbf{F}$ is removed. Find the distance of the particle from the origin after the particle has been in motion for a total of 45 seconds. (5)

(Total 12 marks)

Q22.

A boat moves so that its position vector, $\mathbf{r}$ metres, at time t seconds is given by

$$\mathbf{r} = (4t - 0.01t^2)\mathbf{i} + (5 - 3t - 0.04t^2)\mathbf{j}$$

where the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) State the position of the boat when $t = 0$. (1)
 - (b) Find the time when the boat is due south of its initial position found in part (a). (4)
 - (c) Find the time when the boat is travelling south east. (5)
- (Total 10 marks)

Q23.

A model aeroplane moves in a horizontal plane with a constant acceleration. Initially the aeroplane is at the origin and has velocity $(35\mathbf{i} + 45\mathbf{j}) \text{ m s}^{-1}$. After accelerating for 8 seconds, the velocity of the aeroplane is $(19\mathbf{i} + 13\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

- (a) Show that the acceleration of the aeroplane is $(-2\mathbf{i} - 4\mathbf{j}) \text{ m s}^{-2}$. (3)
 - (b) Find an expression for the position vector of the aeroplane at time t seconds. (3)
 - (c) Find the time when the position vector of the aeroplane is $(300\mathbf{i} + 225\mathbf{j}) \text{ m}$. (7)
- (Total 13 marks)

Q24.

Two forces $\mathbf{F}_1 = (3\mathbf{i} + 4\mathbf{j})\text{N}$ and $\mathbf{F}_2 = (6\mathbf{i} - 8\mathbf{j})\text{N}$, act on a particle. The resultant of these two forces is $\mathbf{F}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

- (a) Find $\mathbf{F}$. (2)
 - (b) Find the magnitude of $\mathbf{F}$. (2)
 - (c) Find the acute angle between $\mathbf{F}$ and the unit vector $\mathbf{i}$. (3)
- (Total 7 marks)

Q25.

A particle is initially at the origin. The particle has a constant acceleration of $(2\mathbf{i} - 3\mathbf{j}) \text{ m s}^{-2}$. Two seconds after it has left the origin the velocity of the particle is $(2\mathbf{i} + 14\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

- (a) Show that the initial velocity of the particle is $(-2\mathbf{i} + 20\mathbf{j}) \text{ m s}^{-1}$. (3)
- (b) Find the distance of the particle from the origin after 3 seconds. (5)
- (c) The speed of the particle is 10 m s^{-1} for the first time T seconds after it has left the origin. Find T . (6)
- (Total 14 marks)

Q26.

A particle has mass 2000 kg. A single force, $\mathbf{F} = 1000t\mathbf{i} - 5000t\mathbf{j}$ newtons, acts on the particle, at time t seconds. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular. No other forces act on the particle.

- (a) Find an expression for the acceleration of the particle. (2)
- (b) At time $t = 0$, the velocity of the particle is $6\mathbf{j} \text{ m s}^{-1}$. Show that at time t the velocity, $\mathbf{v} \text{ m s}^{-1}$, of the particle is given by
- $$\mathbf{v} = \frac{t^2}{4}\mathbf{i} + \left(6 - \frac{5t}{2}\right)\mathbf{j}$$
- (4)
- (c) The particle is initially at the origin. Find an expression for the position vector, $\mathbf{r}$ metres, of the particle at time t seconds. (4)
- (Total 10 marks)

Q27.

A particle moves with a constant acceleration of $(-2\mathbf{i} - 4\mathbf{j}) \text{ m s}^{-2}$. At time $t = 0$ the particle is at the origin and has velocity $(4\mathbf{i} + 7\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

- (a) Find an expression for the position vector of the particle at time t seconds. (2)
- (b) Find the time when the particle has position vector $(3\mathbf{i} + 5\mathbf{j})$ metres. (6)
- (Total 8 marks)

Q28.

A boat moves with a constant acceleration of $(0.2\mathbf{i} + 0.1\mathbf{j}) \text{ m s}^{-2}$, where the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. Time t is measured in seconds.

At time $t = 10$, the boat is at the point A , which has position vector $(30\mathbf{i} - 35\mathbf{j})$ metres.

At time $t = 20$, the boat is at the point B , which has position vector $(70\mathbf{i} - 40\mathbf{j})$ metres.

- (a) Find the vector $\overrightarrow{AB}$. (1)
 - (b) Find the velocity of the boat when $t = 10$. (3)
 - (c) Using your answer to part (b), find the velocity of the boat when $t = 0$. (3)
 - (d) Find the position vector of the boat when $t = 0$. (3)
- (Total 10 marks)**

Q29.

A boat moves so that its position vector, $\mathbf{r}$ metres, at time t seconds is

$$\mathbf{r} = 40 \cos\left(\frac{t}{20}\right) \mathbf{i} + 80 \sin\left(\frac{t}{20}\right) \mathbf{j}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find an expression for the velocity of the boat at time t . (3)
 - (b) In what direction is the boat travelling when $t = 0$? Justify your answer. (2)
 - (c) At what time is the boat travelling due south for the first time? (2)
- (Total 7 marks)**

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Q30.

A particle moves on a horizontal plane subject to the following three forces:

$$\mathbf{F}_1 = (2\mathbf{i} + 3\mathbf{j})\text{N}, \quad \mathbf{F}_2 = (6\mathbf{i} - 8\mathbf{j})\text{N} \quad \text{and} \quad \mathbf{F}_3 = (-2\mathbf{i} + 10\mathbf{j})\text{N}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

Do not include any other forces in your calculations.

- (a) Find the resultant force on the particle. (2)
- (b) The mass of the particle is 8 kg. Find the acceleration of the particle. (1)
- (c) At time $t = 0$, the particle is at the origin and has velocity $(3\mathbf{i} - 2\mathbf{j})\text{ms}^{-1}$.

- (i) Find the velocity of the particle when $t = 40$ seconds. (2)
- (ii) Find the distance of the particle from the origin when $t = 40$ seconds. (5)
- (Total 10 marks)

Q31.

At time $t = 0$, a particle is at the origin and moving with velocity $(4\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-1}$.
When $t = 10$ seconds the position vector of the particle is $(44\mathbf{i} + 28\mathbf{j})$ metres.
The particle is moving with constant acceleration.

- (a) Find the acceleration of the particle. (4)
- (b) Find the position vector of the particle at time t . (3)
- (c) At time $t = T$, the position vector of the particle is $(96\mathbf{i} + 72\mathbf{j})$ metres.
- (i) Find T . (4)
- (ii) Find the speed of the particle at this time. (3)

(Total 14 marks)